

Human Activity Recognition using Smartphone IMU Sensors: CNN-LSTM Deep Learning Model



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Why Human Activity Recognition?

-  Helps in fitness tracking
-  Helps in health monitoring
-  Used in elderly fall detection
-  Used in smart homes and robotics



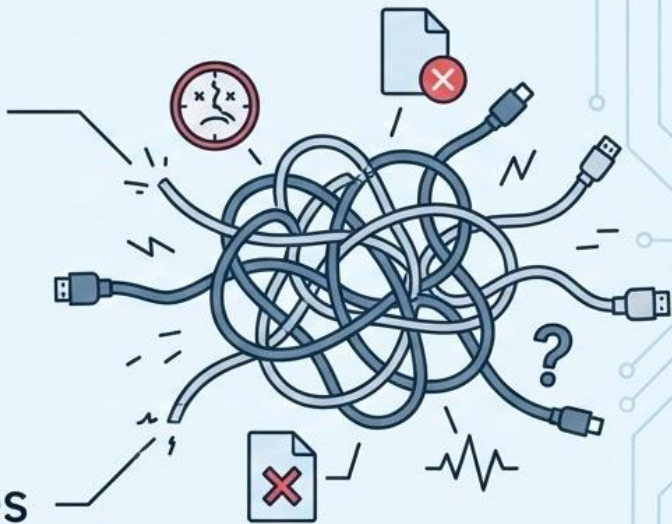
Dataset Collection

- Data collected from your own phone
- Multiple activities recorded
- Each activity folder contains 10 samples
- Each sample has multiple sensors
 - Accelerometer
 - Gyroscope
 - Gravity
 - Orientation
 - Barometer
 - TotalAcceleration
 - Annotation

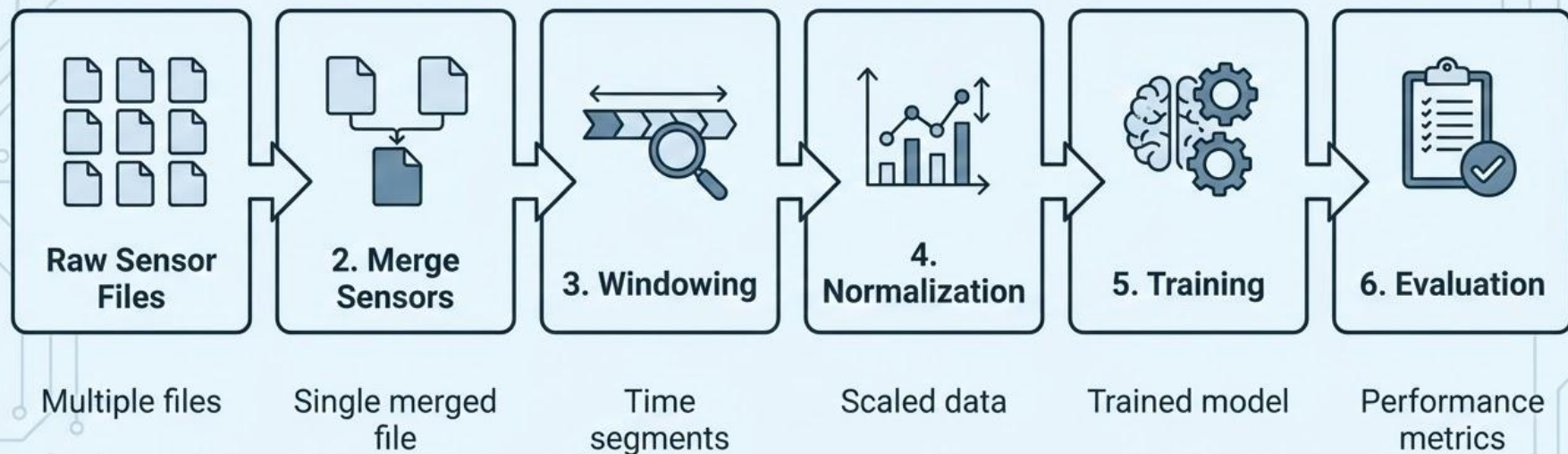


Data Challenge

- Timestamps do not match
- Sensors start/stop at different times
- Different sample lengths
- Missing sensors
- CSVs and XLSX mixed formats
- Columns inconsistent across activities



Solution Architecture



Sensor Merging



Reads all sensor files
Auto-detects CSV/XLSX

Aligns timestamps
using `'merge_asof'`

Creates a single
merged file

Windowing

Different sensors have different scales

Step = 64

Helps model convergence

Avoids exploding gradients

Window size =
= 192 timesteps

Window

Essential for CNN/LSTM stability

 Zero-filling
missing sensors

 All activities
have same
feature count

X.npy
(5963, 192, 26)
(Samples, Time, Features)

y.npy

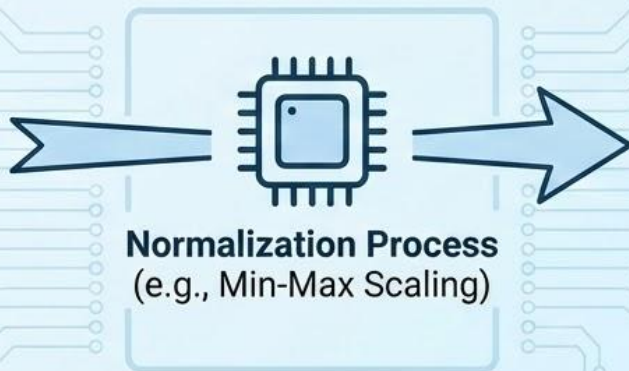
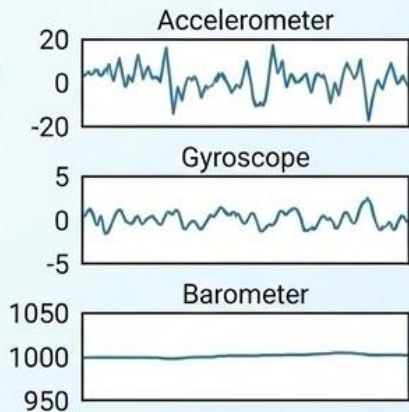
→ labels

labels.npy

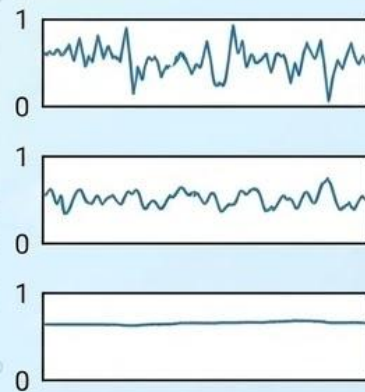
→ activity names

Normalization

Before Normalization:
Different Scales



After Normalization:
Uniform Scale (0-1)



Helps Model

Convergence: Faster, more reliable training.



Avoids Exploding

Gradients: Prevents numerical instability.



Essential for CNN/LSTM

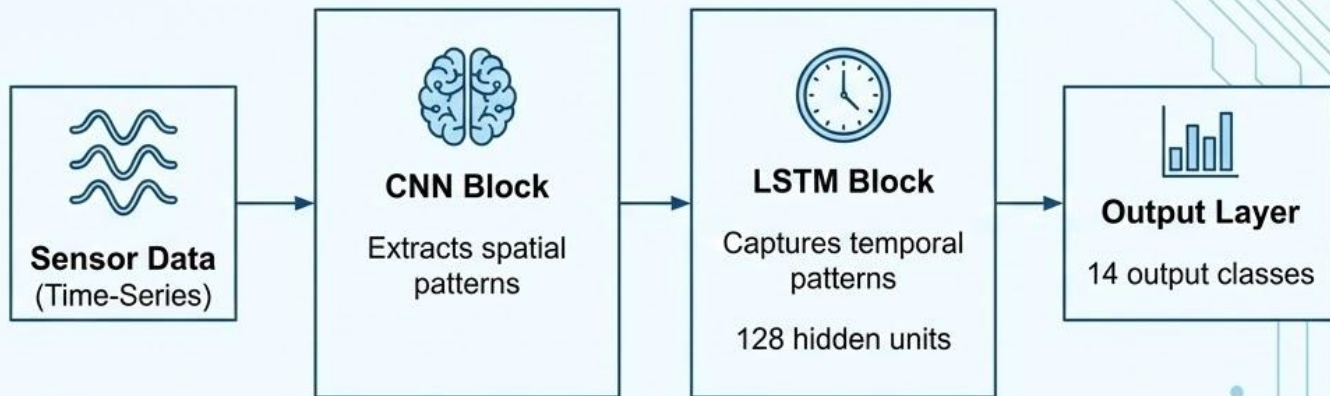
Stability: Ensures deep network performance.

Model Architecture

train_eval.py

Points:

- 20 epochs
- Loss drops correctly
- No overfitting
- GPU/CPU compatible

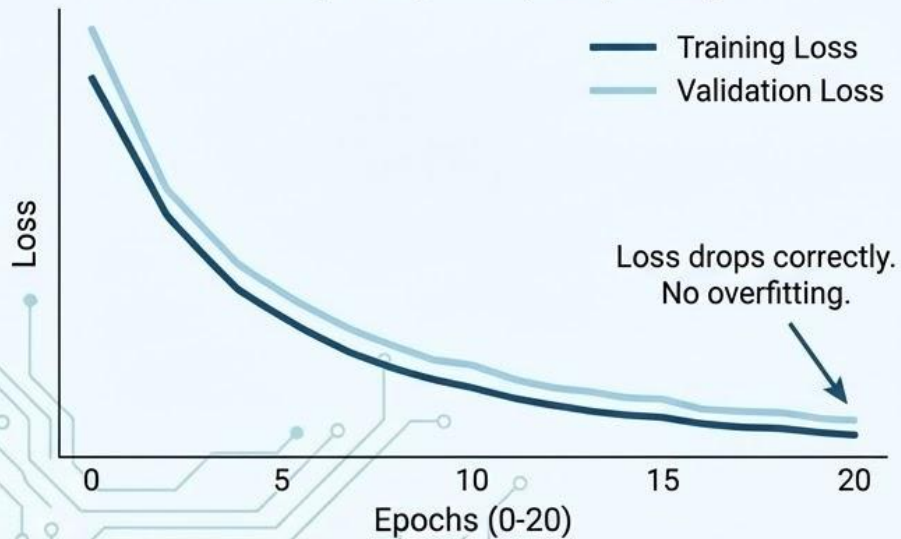


CNN + LSTM Hybrid:
Works best for sensor time-series

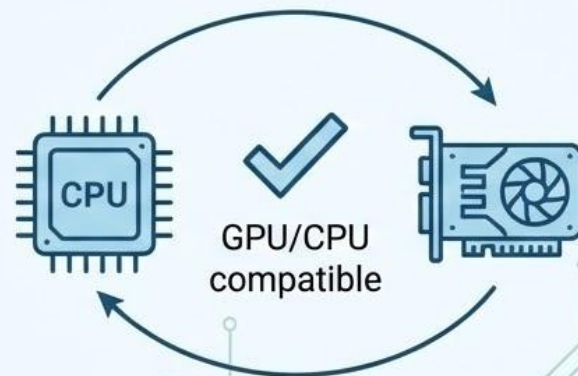
Training

train_eval.py

Training Progress (20 Epochs)

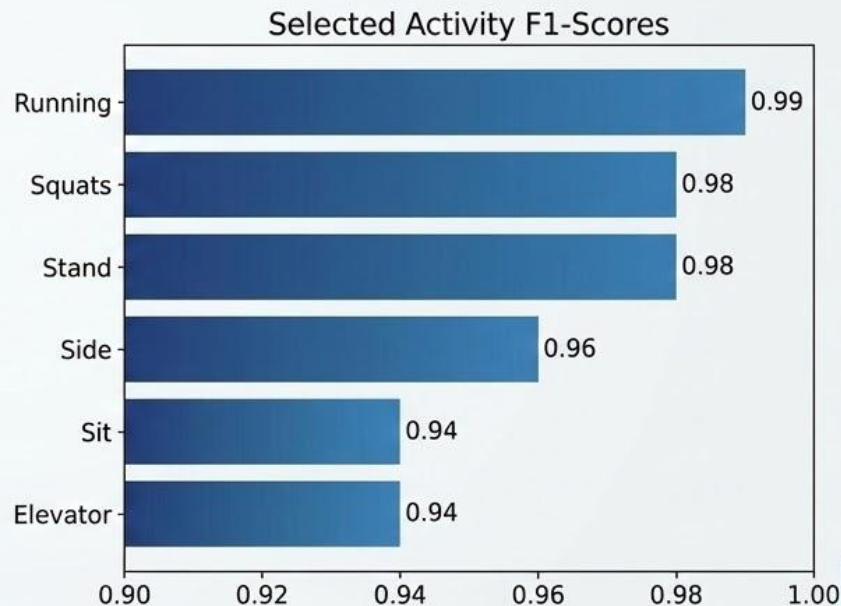


Hardware Compatibility

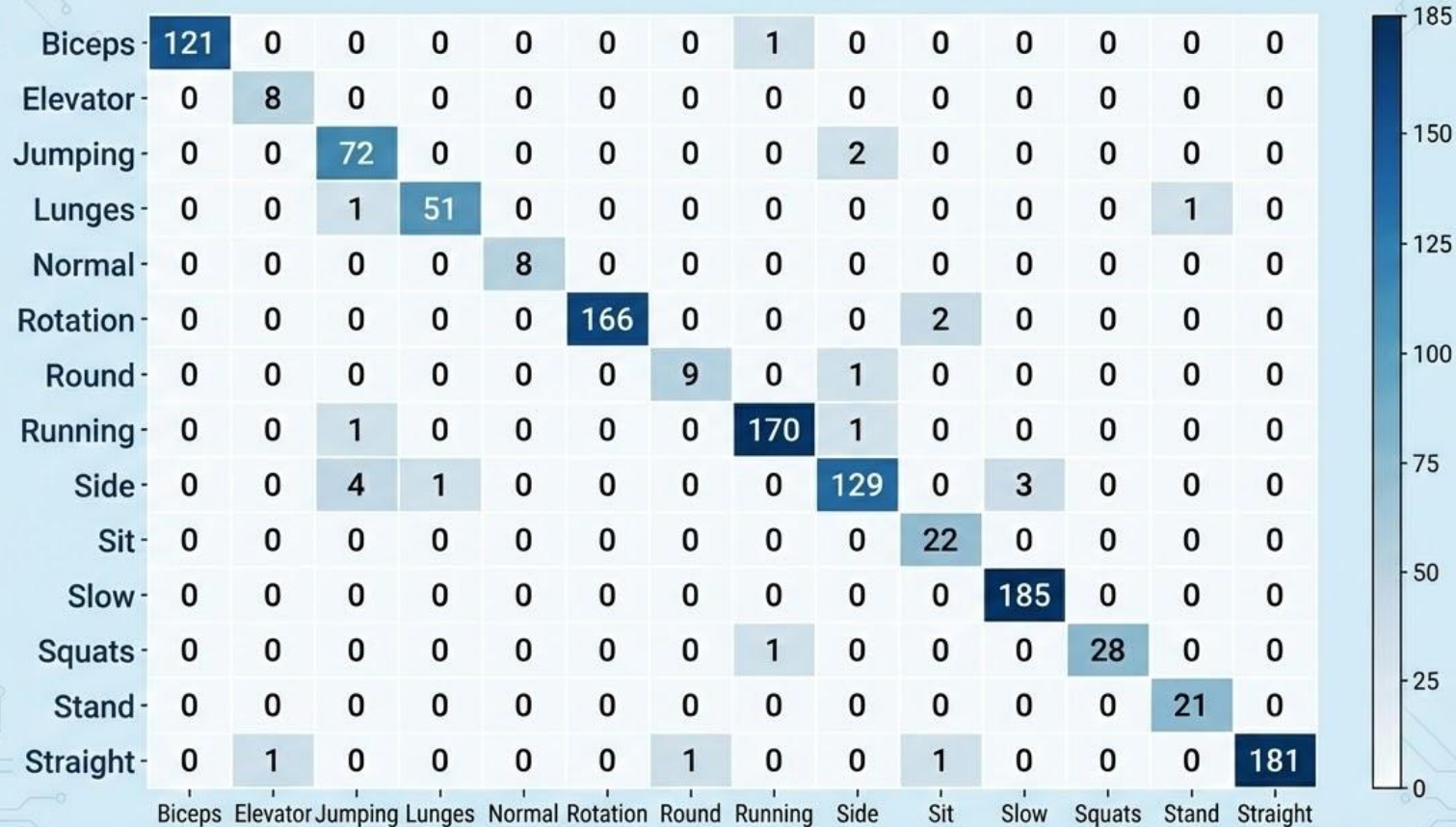


Evaluation Results

- Overall Model Accuracy: **98%**
- Most activities achieve **F1-scores ≥ 0.96** , indicating excellent recognition.
- Minor misclassifications observed between **'Sit'**, **'Elevator'**, and **'Side'** due to similar stationary/semi-stationary sensor patterns.

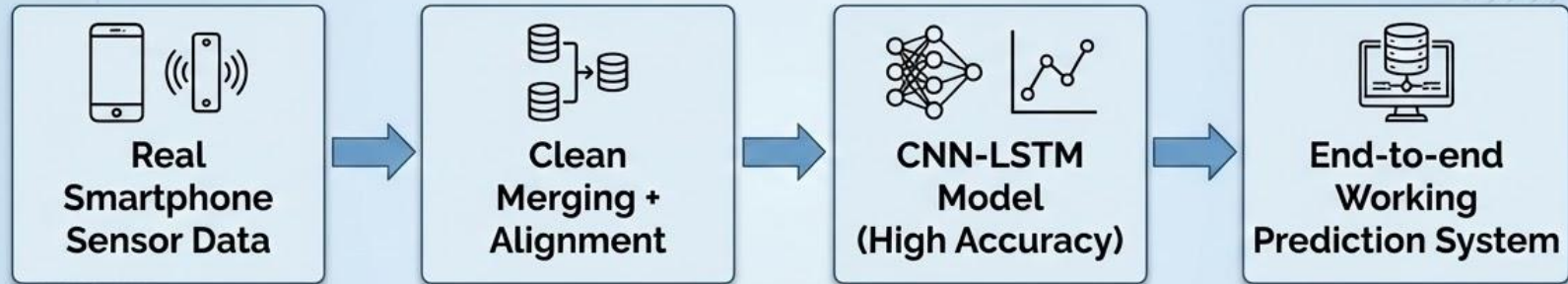


Confusion Matrix



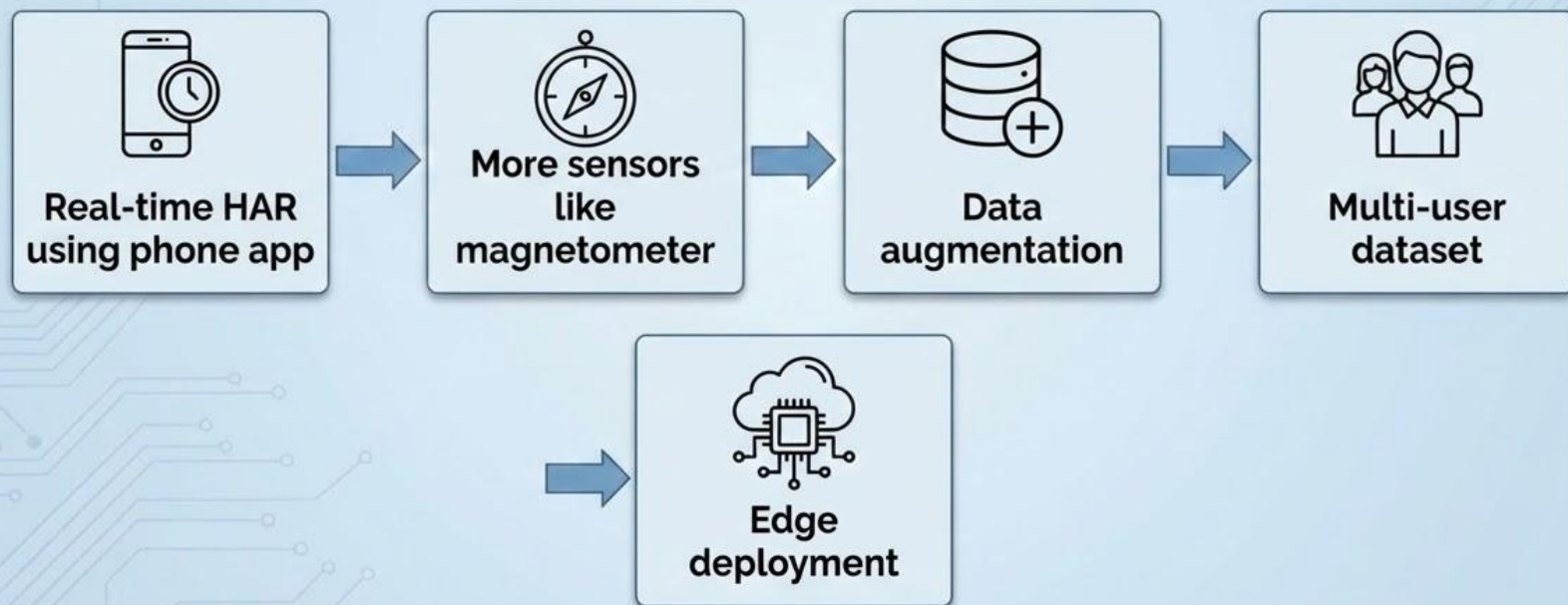
Conclusion

Achievements

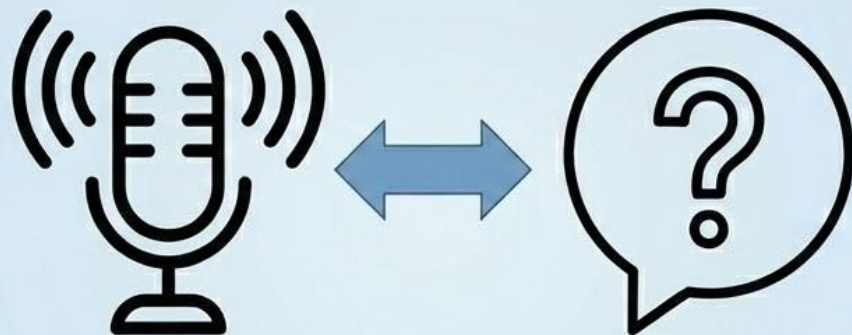


- Built complete HAR pipeline
- Real smartphone sensor data
- Clean merging + alignment
- CNN-LSTM model with high accuracy
- End-to-end working prediction system

Future Work



Q&A



Thank You!

